

A Research Principle for the AI Era

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Research in the AI era should pursue intrinsic value while building the external support needed to sustain that pursuit.

Even when intrinsic value is the sole objective, an initial phase devoted entirely to external return can be necessary for optimal research. The model below makes this statement precise: with a sufficiently long horizon, every optimal strategy begins with full external focus, followed by an optimal phase that combines intrinsic work with the external return required to maintain a growing safety balance.

Here, intrinsic value means the scientific understanding or other substantive contribution that the researcher ultimately values. External return means support convertible into research funding, measured in the same units as expenditure. The distinction separates the purpose of research from the resources that sustain it.

Assumptions

Consider a fixed number n of research cycles. In cycle i , let c_i be expenditure and $p_i \in [0, 1]$ the share allocated to intrinsic work. Write I_i and E_i for intrinsic value and external return, respectively.

1. Linear production and funding renewal. Choose units so that $c_1 = 1$ and

$$I_i = p_i c_i, \quad E_i = b(1 - p_i)c_i, \quad c_{i+1} = E_i.$$

The first two relations give a constant return per unit allocated to either activity. The third makes the next cycle's research budget equal to the current external return. For the final cycle, c_{n+1} is only a convenient name for E_n .

2. A positive and growing safety balance. Let B_i denote the cumulative funding surplus at the end of cycle i :

$$B_0 = 0, \quad B_i = B_{i-1} + E_i - c_i.$$

Require

$$B_i \geq \max\{\delta, \rho B_{i-1}\}, \quad 0 < \delta \leq b - 1, \quad 1 < \rho < b.$$

The fixed floor δ establishes a safety reserve, while ρ requires it to grow. The upper bound on δ makes the first cycle feasible. The inequality $\rho < b$ leaves room to produce intrinsic value while meeting the growth requirement. Funding is assessed at cycle boundaries; the initial expenditure is supplied as seed funding.

3. Intrinsic value as the sole objective. Choose the allocation shares to maximize

$$\sum_{i=1}^n I_i.$$

Returns are deterministic, each cycle's intrinsic value receives equal weight, and external return has value through the funding dynamics and constraints.

4. Enough time for reinvestment. Define

$$K = \min \left\{ m \geq 1 : (b - \rho) \sum_{j=0}^{m-1} \rho^j > 1 \right\}, \quad n > K.$$

Here m is an integer. Since $b > \rho > 1$, K is finite. This assumption gives early external returns enough time to generate later intrinsic value.

Result

Theorem 1. *Under these assumptions, every optimal strategy devotes the first $n - K$ cycles entirely to external return:*

$$p_i = 0 \quad (1 \leq i \leq n - K).$$

There is an optimal strategy that then meets the balance-growth constraint exactly in all remaining cycles:

$$B_i = \rho B_{i-1}, \quad p_i = 1 - \frac{1 + \rho B_{i-1}}{b(1 + B_{i-1})} \in (0, 1) \quad (n - K < i \leq n).$$

Thus, maximizing intrinsic value alone requires an initial phase of full external focus and admits a single subsequent switch to a mixture of intrinsic and external work.

Proof

The funding identity telescopes to

$$B_i = c_{i+1} - 1, \quad c_i = 1 + B_{i-1}.$$

Consequently, at any cycle after the first, a current balance $B \geq \delta$ permits exactly the next balances B' satisfying

$$\rho B \leq B' \leq bB + b - 1.$$

The intrinsic value earned in that cycle is

$$I = 1 - \frac{1}{b} + B - \frac{B'}{b}.$$

The upper endpoint corresponds to $p = 0$; the lower endpoint meets the required balance growth exactly.

Let $V_m(B)$ be the largest total intrinsic value attainable with m cycles remaining and current balance $B \geq \delta$. We show by induction that V_m is affine in B , and denote its slope by A_m . Since $V_0 = 0$, $A_0 = 0$. The recursion is

$$V_m(B) = 1 - \frac{1}{b} + B + \max_{\rho B \leq B' \leq bB + b - 1} \left\{ V_{m-1}(B') - \frac{B'}{b} \right\}.$$

If V_{m-1} is affine, the coefficient of B' is $A_{m-1} - 1/b$. Therefore the maximizing choice is the upper endpoint when $A_{m-1} > 1/b$, and the lower endpoint when $A_{m-1} < 1/b$. At equality, either endpoint is optimal; choose the lower one. Substitution shows that V_m is affine and gives

$$A_m = \begin{cases} 1 - \rho/b + \rho A_{m-1}, & A_{m-1} \leq 1/b, \\ b A_{m-1}, & A_{m-1} > 1/b. \end{cases}$$

Until the threshold is crossed,

$$A_m = \frac{b - \rho}{b} \sum_{j=0}^{m-1} \rho^j.$$

Hence $A_{K-1} \leq 1/b < A_K$. After this crossing, the second branch keeps $A_m > 1/b$. It follows that every cycle with more than K cycles remaining has the unique optimal choice of full external focus. With at most K cycles remaining, choosing the lower endpoint is optimal.

For the first cycle, the feasible interval is $\delta \leq B_1 \leq b - 1$. Its objective has coefficient $A_{n-1} - 1/b$ on B_1 . Since $n > K$, this coefficient is strictly positive, so every optimum chooses $B_1 = b - 1$, equivalently $p_1 = 0$. The same argument applies to the subsequent cycles with more than K cycles remaining.

In the last K cycles, choose $B_i = \rho B_{i-1}$. The formula for p_i follows from $E_i = 1 + B_i = b(1 - p_i)(1 + B_{i-1})$. Since $B_{i-1} > 0$ and $1 < \rho < b$, this share lies strictly between zero and one. This proves the result. $\square$

Build external support early so that the full research trajectory can deliver more intrinsic value.